

This video elucidates key engineering mathematics concepts regarding random variables, focusing on the distinction between discrete and continuous systems, the application of calculus in probability, and methodologies for solving GATE-level problems.

Summary

The transition from discrete to continuous random variables requires a shift from algebraic summation to integral calculus.

Core principles:

- Probability theory relies on two fundamental functions:
 - Probability Mass/Density Function
 - Cumulative Distribution Function (CDF)
- For continuous variables, probability at any specific point is zero.
- Probability is defined over intervals as the area under the density curve.
- Unknown constants in density functions are determined by normalizing the total area to

1. Classification of Random Variables

The behavior of a random variable depends on its domain.

Feature	Discrete Random Variable (DRV)	Continuous Random Variable (CRV)
Domain Type	Integer values / “Arrivals”	Intervals or Infinite sets
Countability	Countable	Uncountable / Infinite
Core Function	Probability Mass Function (PMF)	Probability Density Function (PDF)
Mathematical Tool	Summation	Calculus (Integration)
Point Probability	Defined as $P(X = x)$	Always zero: $P(X = x) = 0$

1.1 Discrete Random Variables (DRV)

- Defined for integer values.
- PMF gives probability at a specific point.
- CDF $F(x)$ represents cumulative probability up to x :
Sum of probabilities of all outcomes $\leq x$.

1.2 Continuous Random Variables (CRV)

- Defined over intervals.
- Domain contains uncountable points.
- Probability at a single point is zero.

For any interval $[a, b]$:

$$P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b) = P(a < X \leq b)$$

2. Fundamental Probability Functions

2.1 Probability Density Function (PDF)

The PDF $f(x)$ describes the likelihood of a continuous random variable within a range.

Total probability condition:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

2.2 Cumulative Distribution Function (CDF)

The CDF $F(x)$ represents:

$$F(x) = P(X \leq x)$$

For continuous variables:

$$F(x) = \int_{-\infty}^x f(t) dt$$

3. Key Mathematical Principles and Problem-Solving

3.1 Normalization to Find Constants

To find an unknown constant (a, c, k) in a PDF:

1. Integrate the function over its defined limits.
2. Set the result equal to 1.

Example:

If

$$f(x) = \frac{a}{x^{\mu}}, \quad x > 1, \quad \mu > 1$$

Then:

$$\int_1^{\infty} \frac{a}{x^{\mu}} dx = 1$$

Result:

$$a = \mu - 1$$

3.2 Interval Probability

To compute probability between a and b:

$$P(a < X < b) = \int_a^b f(x) dx$$

3.3 Conditional Probability in Continuous Domains

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Number Line Technique Example

Find: $P(X \leq 2 \mid X \geq 1)$

1. Event A ($X \leq 2$): $(-\infty, 2]$
2. Event B ($X \geq 1$): $[1, \infty)$
3. Intersection ($A \cap B$): $[1, 2]$

Thus:

$$[P(X \leq 2 \mid X \geq 1) = \frac{\int_1^2 f(x) dx}{\int_1^{\infty} f(x) dx}]$$

4. Discrete Probability and Series Applications

For discrete variables following a pattern (e.g., even numbers 2, 4, 6, ...), probabilities may form a Geometric Progression (GP).

Example:

$$[P(X = x) = \left(\frac{2}{3}\right)^x \quad \text{for even } x]$$

Series becomes:

$$[\left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^4 + \left(\frac{2}{3}\right)^6 + \dots]$$

This is a GP:

$$[S = \frac{a}{1 - r}]$$

where

a = first term

r = common ratio

5. Visual Representation of Variable Domains

Discrete Domain (Point-Based)

Points:

1 2 3 4
• • • •

Probabilities exist at individual dots.

Continuous Domain (Interval-Based)

Interval: [-----|-----|-----]
1 x 1+h 2

Moving from 1 to 1+h creates an uncountable neighborhood where point probability becomes zero.